21 Sketch the graph of

$$y = 2x^2 - 8x - 5$$

showing the coordinates of the turning point and the exact coordinates of any intercepts with the coordinate axes.

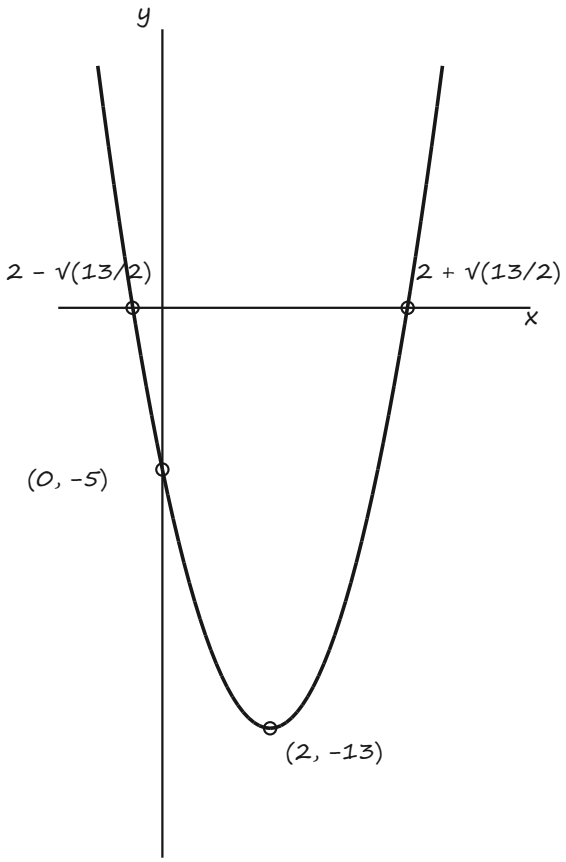
$y = 2x^2 - 8x - 5 = 2(x - 2)^2 - 13.$

Turning point: $(2, -13)$; y-intercept: $(0, -5)$.

For $y = 0$, $(x - 2)^2 = 13/2$,

so x-intercepts: $(2 \pm \sqrt{(13/2)}, 0)$.

5 marks total
Complete square / start roots
Exact roots
Turning point
y-intercept
Correct labelled sketch



(Total for Question 21 is 5 marks)

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